

D6. A distillation column operating at 1.0 atm is separating a mixture of methanol and water. The feed is a saturated liquid. The column has a total condenser and a partial reboiler. The reflux is returned as a saturated liquid, and constant molal overflow is assumed to be valid. Feed flow rate is 50 kmol/h. We desire a bottoms composition of 5 mol% methanol and a distillate composition of 90 mol % methanol. The reflux ratio is $L/D = 1$, and the boilup ratio is $\bar{V}/B = 2$. VLE data are in Table 2-8.

- Derive and plot the operating lines.
- Starting at the top of the column, determine the optimum feed stage and the total number of equilibrium stages needed.
- Determine the feed mole fraction methanol.

a) $V_y = Lx + Dx_D$ $y = \left(\frac{L}{V}\right)x + \left(1 - \frac{L}{V}\right)x_D$
 $V = L + D$

Slope: $\frac{L}{V} = \frac{L}{L+D} = \frac{L/D}{L/D+1} = \frac{1}{1+1} = \frac{1}{2}$

y intercept: $\left(1 - \frac{L}{V}\right)x_D = \left(1 - \frac{1}{2}\right)(0.90) = 0.45$

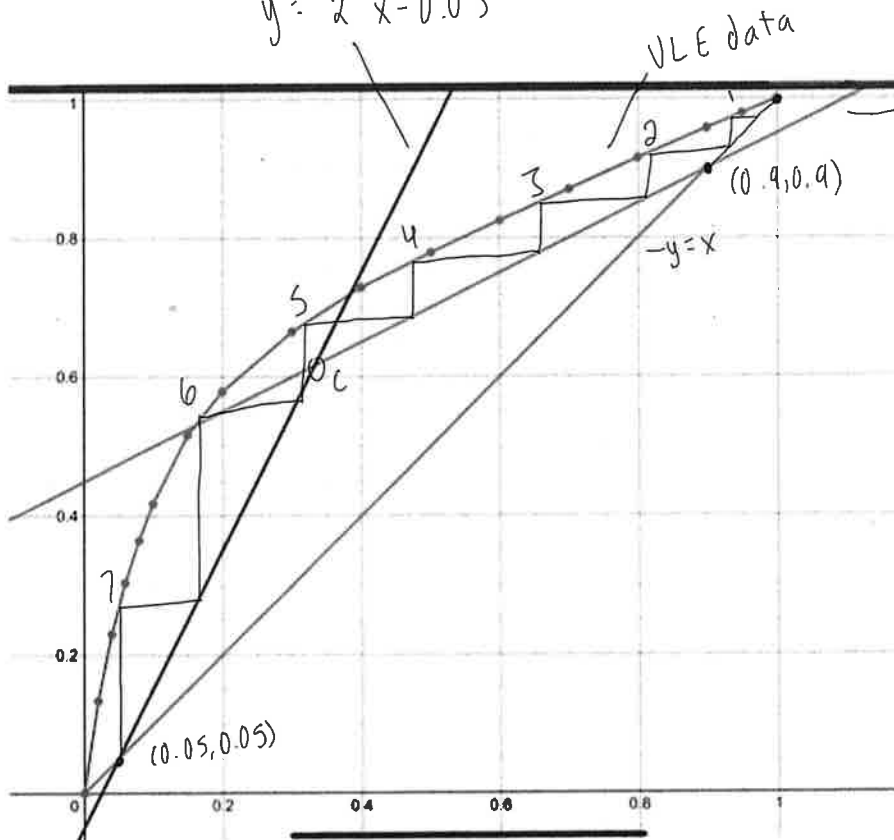
y = x curve: $y_1 = x_0 = x_D = 0.9 \rightarrow (0.9, 0.9)$

$y = 0.5x + 0.45$

$\bar{L} = \bar{V} + B$ $y = \frac{\bar{V}}{B}x - \left(\frac{\bar{V}}{B} - 1\right)x_B$

slope: $\frac{\bar{V}}{B} = 2$ y-intercept: $\left(\frac{\bar{V}}{B} - 1\right)x_B = (2 - 1)(0.05) = 0.05$

$y = 2x - 0.05$



b) optimum feed stage:

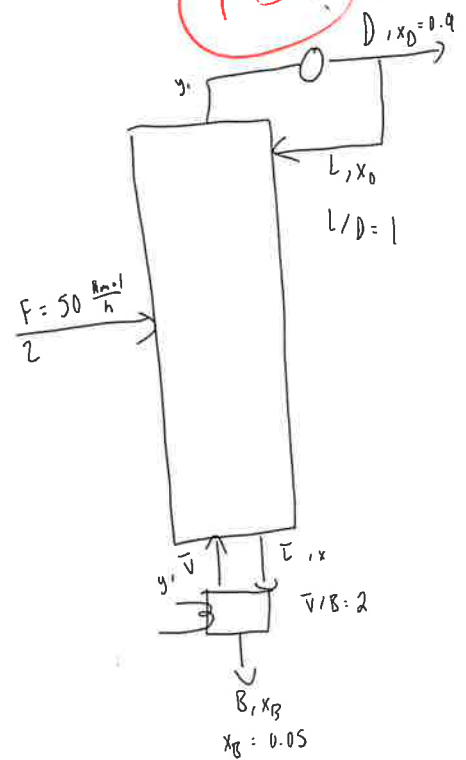
5

total # stages: 7

c) $x = 0.34$

*intersection

0.475



2

17. A distillation column with a total condenser is separating a feed that is 30 mol% benzene from ethanol. Distillate $x_D = 0.95$, and bottoms $x_B = 0.15$, both mole fractions. Assume CMO is valid, $L/V = 0.8$ (sharp ratio $\alpha = 8$), and $B = 1.0$. Find the composition of the liquid leaving the 6th stage below the total condenser.

b. The column has a partial reboiler that acts as an equilibrium contact. Find the vapor composition leaving the second stage above the partial reboiler.

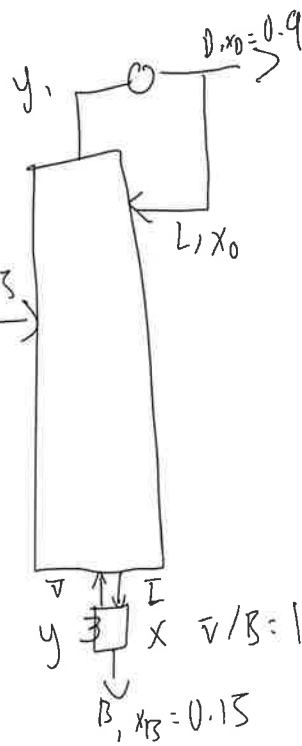
c. Find optimum feed plate location, total number of stages, and required q value of feed.

Equilibrium data for acetone (4) and ethanol at 1 atm (Perry et al., 1993, pp. 11-46):

x_A	0	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.50	0.60	0.70	0.80	0.90	1.0
y_A	0	0.262	0.348	0.417	0.478	0.524	0.566	0.604	0.647	0.704	0.802	0.904	0.979	1.0
$T, ^\circ C$		77.1	74.5	72.9	71.6	70.4	69.3	68.2	66.4	64.1	60.2	55.0	51.0	

$$L/V = 0.8$$

$$\bar{V}/B = 1$$



$$V y = L x + D x_D \rightarrow y = \frac{L}{V} x + \frac{D}{V} x_D$$

$$V = L + D \rightarrow \frac{D}{V} = \frac{V}{V} - \frac{L}{V} \rightarrow y = \frac{L}{V} x + (1 - \frac{L}{V}) x_D$$

$$y = 0.8x + (1 - 0.8)(0.9) = 0.8x + 0.18$$

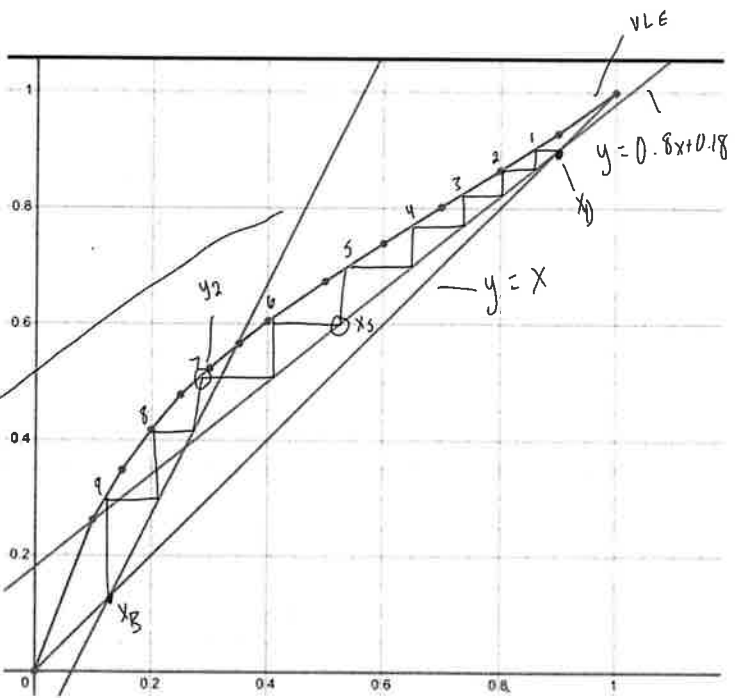
$$y = x; y_1 = x_D = x_0 = 0.9 \quad (0.9, 0.9)$$

$$y = \frac{\bar{L}}{\bar{V}} x - (\frac{\bar{L}}{\bar{V}} - 1) x_B$$

$$\text{slope: } \frac{\bar{L}}{\bar{V}} = \frac{\bar{V}/B + 1}{\bar{V}/B} = \frac{1+1}{1} = 2$$

$$y\text{-intercept: } (\frac{\bar{L}}{\bar{V}} - 1) x_B = (2 - 1)(0.15) = 0.15$$

$$y = 2x - 0.15$$



a) $x_5 = 0.53$

b) $y_2 = 0.52$

c) optimum feed plate location: 7
stages: 9

$$q = 0.67$$

$$0.8x + 0.18 = 2x - 0.15$$

$$1.2x = 0.31 \quad y = 0.8x + 0.18$$

$$x = 0.258 \rightarrow y = 0.3864$$

$$\text{slope: } \frac{y_2 - y_1}{x_2 - x_1} = \frac{0.3864 - 0.3}{0.258 - 0.3} = -2.06$$

$$\frac{q}{q-1} = -2.06 \rightarrow q = -2.06(q-1)$$

$$q = -2.06q + 2.06$$

$$3.06q = 2.06 \quad q = 0.67$$

D11. A beer still (see Figure 3-9 in Problem 3.E2) is being used to separate ethanol from water at 1.0 atm. The saturated liquid feed flow rate is $F = 840.0$ kmol/h. The feed is 44.0 mol% ethanol. The saturated vapor steam is pure water with ratio of steam flow rate S to feed rate, $S/F = 2/3$. We desire a bottoms product that is 4.0 mol% ethanol. CMO is valid. Equilibrium data are in Table 2-1.

- Find the mole fraction of ethanol in the distillate vapor, y_D .
- Find the number of equilibrium stages required.
- If the feed is unchanged and the S/F ratio is unchanged, but the number of stages is increased to a very large number, what is the lowest bottoms mole fraction of ethanol that can be obtained?

$$F + S = D + B \rightarrow D = F + S - B$$

$$F z_F + S y_S = D y_D + B x_B$$

CMO: $F = B = L, D = S = V$

$$\frac{S}{F} = \frac{2}{3} \rightarrow S = \frac{2}{3}(840) = 560 \frac{\text{kmol}}{\text{h}}$$

$$a) F z_F = D y_D + B x_B \rightarrow (840)(0.44) = 560 y_D + (840)(0.04)$$

$$336 = 560 y_D \quad y_D = 0.6$$

$$b) y = \frac{B}{S} x + (1 - \frac{B}{S}) y_D = \frac{840}{560} x + (1 - \frac{840}{560})(0.6)$$

$$y = 1.5x - 0.3 \quad (\text{top})$$

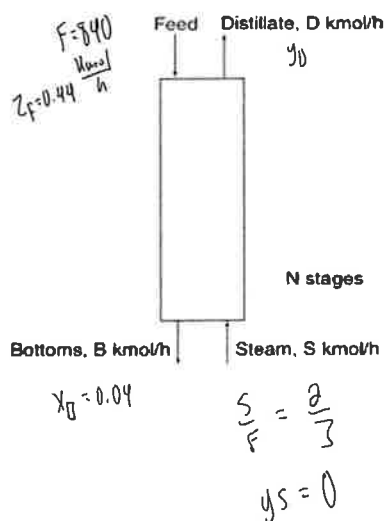
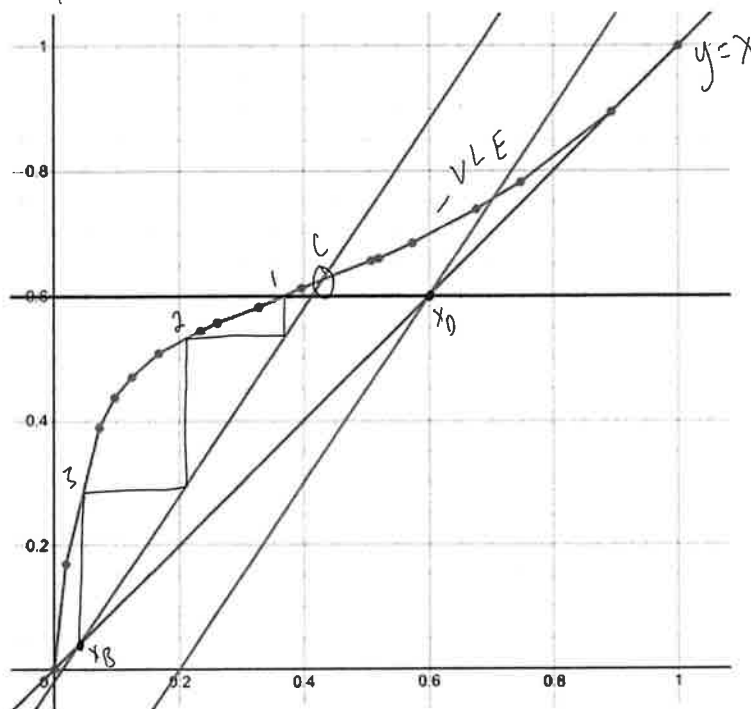
$$y = \frac{B}{S} x + (1 - \frac{B}{S}) x_B = \frac{840}{560} x + (1 - \frac{840}{560})(0.04)$$

$$y = 1.5x - 0.02$$

$$\text{feed line: } q=1 (\text{sat liq}) : x=z=0.44$$

stages:

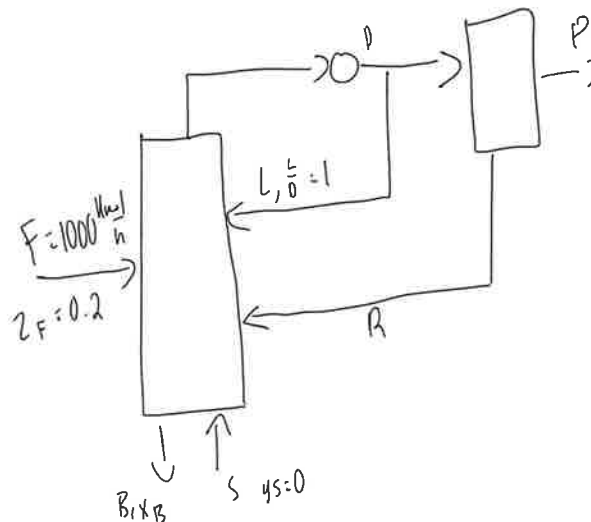
3



D12. A distillation column is connected to another separation device. The fresh feed to the column is $F = 1000.0 \text{ kmol/h}$ of a saturated vapor that is 20.0 mol% ethanol and 80.0 mol% water. The feed is input at its optimum feed stage. The distillation column has a partial condenser and no reboiler. The column is heated with direct injection of pure steam that is a saturated vapor at 1.0 atm. The ratio bottoms flow rate/steam flow rate $= B/S = 5.0$. The reflux ratio $L_0/D = 1.0$. The distillate from the column is 70.0 mol% ethanol and the bottoms is 1.0 mol% ethanol. The column operates at a pressure of 1.0 atm, the reflux is returned as a saturated liquid, and CMO is assumed to be valid. VLE data are in Table 2-1.

The distillate is sent to a separation device that, because it operates on a different principle than distillation, is not limited by the presence of an azeotrope. The device produces product P and recycle stream R that is a saturated liquid returned to the distillation column at its optimum feed stage. The product P concentration from the device is 98.7 mol% ethanol.

- a. Find steam flow rate S, bottoms flow rate B, distillate flow rate D, product flow rate P, and recycle rate R—all in kmol/h—and find mole fraction of



ethanol in recycle stream x_R .

- b. For the distillation column, find the optimum feed plate for the recycle stream, the optimum feed plate for the fresh feed, and the total number of equilibrium stages required.

a) $F = 1000 \frac{\text{kmol}}{\text{h}}$ $z_F = 0.2$ $F \text{ is } 0.4$
 $\frac{B}{S} = 5$ $\frac{L_0}{D} = 1$ $x_D = 0.7$ $x_B = 0.01$ $y_S = 0$ $x_P = 0.987$
 Overall: $F + R + S = D + B \rightarrow D = F + R + S - B$ $B = 5S$
 $F z_F + R x_R + S y_S = D x_D + B x_B$

Separate:

$$D = P + R$$

$$D x_D = P x_P + R x_R \rightarrow (P + R) x_D = P x_P + R x_R$$

$$P x_D + R x_D = P x_P + R x_R$$

$$P(x_D - x_P) = R(x_R - x_D)$$

$$F + R + S = (P + R) + B \rightarrow F + S = P + B$$

$$P = F + S - B = F - 4S$$

$$F z_F + R x_R = (P + R) x_D + 5S x_B \quad (B)$$

$$R(x_R - x_D) = P x_D + 5S x_B - F z_F$$

$$P(x_D - x_P) = P x_D + 5S x_B - F z_F$$

$$-P x_P = 5S x_B - F z_F$$

Table 2-1 Vapor-liquid equilibrium data for ethanol and water at 1 atm and 1.0 mole fractions

x_{eth}	y_{eth}	T_{eth}	P_{eth}	$T, ^\circ\text{C}$
0	1.0	0	1.0	100
0.019	0.981	0.170	0.980	95.5
0.0721	0.9279	0.3891	0.6109	89.0
0.0966	0.9034	0.4375	0.5625	86.7
0.1238	0.8762	0.4704	0.5296	85.3
0.1661	0.8334	0.5089	0.4911	84.1
0.2137	0.7661	0.5445	0.4555	82.7
0.2608	0.7302	0.5580	0.4420	82.3
0.3271	0.6727	0.5826	0.4174	81.5
0.3965	0.6018	0.6122	0.3878	80.7
0.4079	0.4921	0.6164	0.3816	79.8
0.4198	0.4892	0.6199	0.3801	79.7
0.4332	0.4268	0.6841	0.3150	79.1
0.4763	0.3237	0.7385	0.2615	78.74
0.7472	0.2528	0.7815	0.2185	78.41
0.9043	0.1057	0.8941	0.1057	78.15
1.00	0	1.00	0	78.30

$$P(x_D - x_P) = F(x_D - z_F) + S(5x_B - 4x_D)$$

$$(F - 4S)(x_D - x_P) = F(x_D - z_F) + S(5x_B - 4x_P)$$

$$F(x_D - x_P) - 4S(x_D - x_P) = F(x_D - z_F) + S(5x_B - 4x_D)$$

$$S[-4(x_D - x_P) - (5x_B - 4x_D)] = F[(x_D - z_F) - (x_D - x_P)]$$

$$S(4x_P - 5x_B) = F(x_P - z_F)$$

$$S = \frac{F(x_P - z_F)}{4x_P - 5x_B} = \frac{(1000)(0.987 - 0.2)}{4(0.987) - 5(0.01)} = 201.9 \frac{\text{kmol}}{\text{h}}$$

$$B = 5S = 5(201.9) = 1009.5 \frac{\text{kmol}}{\text{h}}$$

$$P = F - 4S = 1000 - 4(201.9) = 192.4 \frac{\text{kmol}}{\text{h}}$$

$$D = \frac{S + F}{2} = \frac{201.9 + 1000}{2} = 600.95 \frac{\text{kmol}}{\text{h}}$$

$$R = D - P = 600.95 - 192.4 = 408.55 \frac{\text{kmol}}{\text{h}}$$

$$x_R = \frac{Dx_D - Px_P}{R} = \frac{(600.95)(0.7) - (192.4)(0.987)}{408.55} = 0.56$$

$$b) V = L + D$$

$$\frac{L}{V} = \frac{L}{L+D} = \frac{L/D}{L/D+1} = \frac{1}{2}$$

$$\frac{D}{L+D} x_D = \frac{1}{L/D+1} (0.7) = 0.35$$

$$y = 0.5x + 0.35 \quad (\text{top})$$

$$y = \frac{L}{S}x - \frac{B}{S}x_B$$

$$(MO: L = D)$$

$$y = \frac{600.95}{201.9}x - \frac{1009.5}{201.9}(0.01)$$

$$y = 2.98x - 0.05 \quad (\text{bottom})$$

feed line:

$$q=0 \rightarrow y = 0.2$$

recycle line:

$$q=1 \rightarrow x = x_R = 0.56$$

optimum feed plate recycle: 2

optimum feed plate feed: 6

stages: 6

